

Filecoin Network Supply Forecast: Scenario Analysis

CEL — February 2026

Executive Summary

This document presents a two-year forward simulation of the Filecoin network under two scenarios that vary storage provider (SP) onboarding and renewal behavior.

The central finding is that **even holding current conditions flat, locked FIL and L/CS decline over the forecast horizon**. The Decline scenario, which extrapolates recent trends in onboarding and renewal behavior, projects a more significant contraction: locked FIL falls 39% while circulating supply grows 15%, compressing L/CS from 0.122 to 0.065. **Based on the trajectory of the past year, the Decline scenario is the more realistic projection.**

Table 1: Forecast summary. “Current” reflects on-chain values as of February 21, 2026. Status Quo and Decline columns show values at the end of the two-year forecast horizon.

Metric	Current	Status Quo	Decline
Network Locked (M FIL)	101.8	99.4	62.5
Circulating Supply (M FIL)	836.6	924.5	960.0
Locked / Circ Supply	0.122	0.108	0.065
Pledge per 32 GiB Sector (FIL)	0.172	0.162	0.360
1Y Forward FoFR (%)	22.7	21.9	50.2

Implications and Recommendations

1. **Circulating supply is approaching 1 billion FIL.** Under both scenarios, circulating supply crosses 900M FIL within the forecast horizon and approaches 960M under Decline. With locked FIL contracting and minting ongoing, the network is on track to cross the 1B threshold. Once crossed, there is no protocol mechanism to reverse this — the only path back is increased locking demand.
2. **The network needs an alternative locking mechanism urgently.** Sector pledge is the only meaningful source of FIL locking today, and it is shrinking. Implementing **veFIL** (or an

equivalent vote-escrowed locking mechanism) would create a second source of locking demand that is not tied to storage onboarding. This should be treated as a high-priority initiative.

3. **It is worse for Filecoin if no one wants to onboard capacity than if there are no paid deals.** The onboarding decline is the more existential threat. Paid storage deals are desirable, but the network’s economic security depends on SPs continuing to onboard and renew sectors regardless of deal status. Policy decisions should prioritize removing barriers to onboarding above all else.
4. **FIL+ should be reconsidered.** Filecoin no longer provides an effective subsidy to SPs — even with paid deals, the economics are marginal. FIL+ has become an administrative burden across the ecosystem (notaries, allocators, compliance) without delivering commensurate onboarding growth. We should strongly consider removing FIL+ entirely by granting all sectors the 10x quality multiplier by default. This would eliminate friction for SPs, reduce governance overhead, and acknowledge the reality that the verified-deal distinction no longer drives meaningful network growth.

Scenario Definitions

The simulation is initialized with 365 days of on-chain history (February 2025 – February 2026). Key inputs are derived from 30-day trailing statistics.

Current network parameters (30-day):

- Raw byte onboarding: 0.99 PiB/day (median)
- Renewal rate: 72.1% (mean)
- FIL+ rate: 97.1% (median)

Two scenarios are modeled:

Status Quo — Current onboarding and renewal rates are held flat for the duration of the forecast. This represents the optimistic case where current conditions do not deteriorate further.

Decline — New onboarding decays exponentially at 3x per year (falling to one-third of its current level after 12 months and one-ninth after 24 months), while the renewal rate declines linearly from 72% to 36% over the forecast horizon. This scenario extrapolates the declining trends observed over the past year in both SP onboarding and renewal behavior. Raw byte onboarding has already declined approximately 3x over the past 12 months, and a continued deterioration in SP participation would compound through both reduced new entrants and increasing non-renewal of existing capacity.

Table 2: Scenario input parameters.

Parameter	Status Quo	Decline
RB Onboarding	0.99 PiB/d (flat)	0.99 → 0.33 (1Y) → 0.11 (2Y) PiB/d

Parameter	Status Quo	Decline
Renewal Rate	72.1% (flat)	72% $\rightarrow$ 36% (linear)
FIL+ Rate	97.1% (flat)	97.1% (flat)

Forecast Results

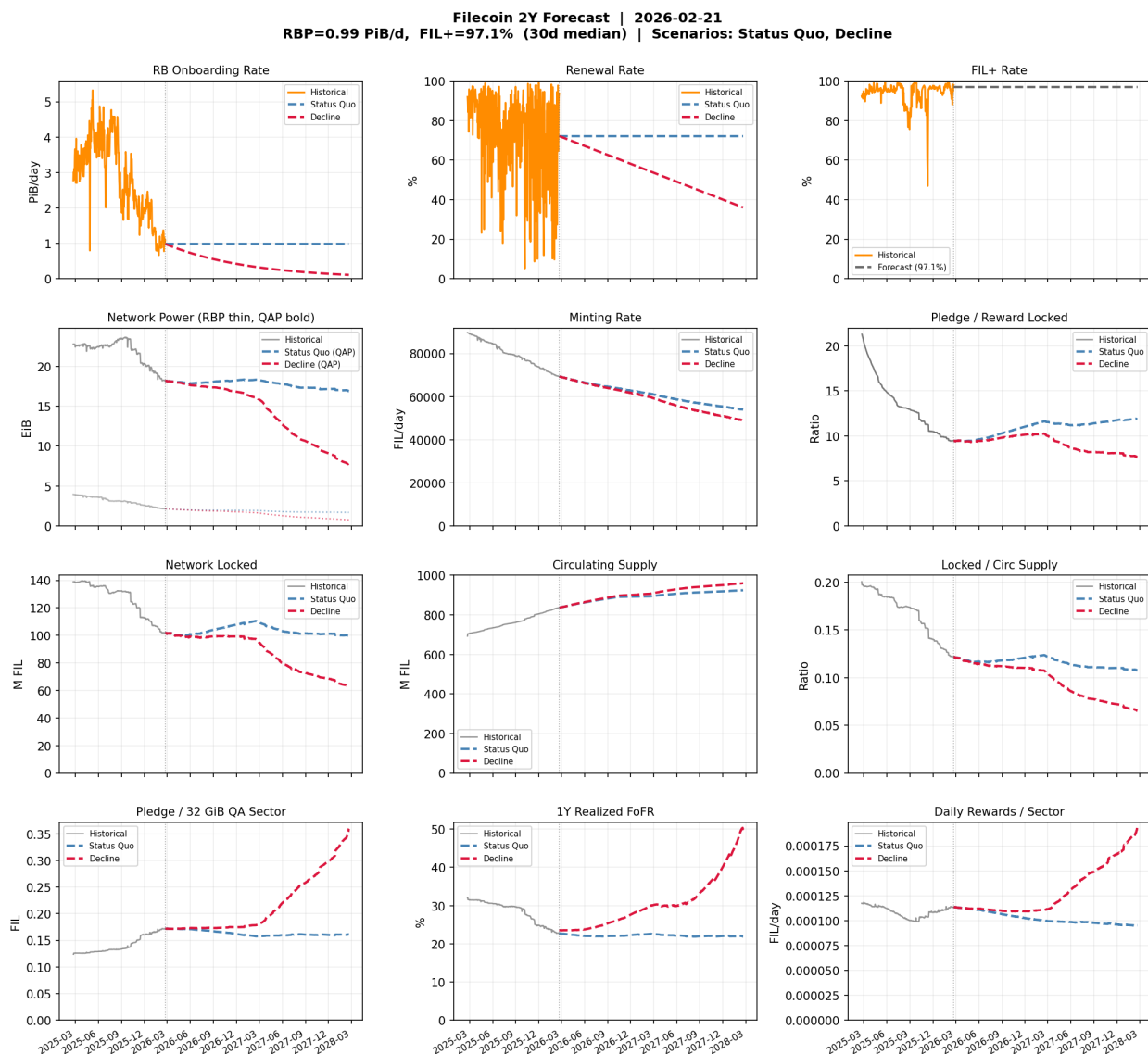


Figure 1: Two-year scenario forecast panel. Rows from top: inputs, network power and minting, supply metrics, and SP economics. The vertical dotted line marks the transition from historical data (solid gray) to the forecast period (dashed, color-coded by scenario).

Network Power

Under the Status Quo scenario, quality-adjusted power (QAP) declines modestly from 18.2 EiB to approximately 16.9 EiB (-7%). At current onboarding rates, new capacity does not fully offset sector expiry, so the network slowly contracts even with flat inputs. The 9.7x FIL+ multiplier amplifies the effect of raw byte changes on QAP, so even modest differences in renewal behavior produce significant QAP movements.

The Decline scenario shows QAP falling to approximately 7.5 EiB (-59%) as both onboarding and renewals contract. The exponential decay in new onboarding reduces the flow of new capacity, while the declining renewal rate accelerates the release of expiring sectors.

Locked FIL and Circulating Supply

Locked FIL is the most consequential metric for circulating supply dynamics.

Under the **Status Quo**, locked FIL declines modestly from 101.8M to 99.4M (-2%). Circulating supply grows from 837M to 925M (+11%), driven by ongoing minting and vesting. The net result is that L/CS declines from 0.122 to 0.108 (-11%).

Under the **Decline** scenario, the trajectory unfolds in two distinct phases:

Table 3: Decline scenario — cumulative change from forecast start.

Horizon	QAP	Locked FIL	Circ Supply	L/CS
Start	18.2 EiB	101.8M	836.6M	0.122
+6 months	-5%	-3%	+6%	-8%
+1 year	-12%	-5%	+9%	-13%
+18 months	-41%	-28%	+12%	-36%
+2 years	-59%	-39%	+15%	-47%

- **First year (inertia phase):** Although QAP begins to contract more noticeably (-12%), locked FIL is only down 5%. This reflects the 360-day sector duration: pledge locked by renewing sectors today is not released until those sectors expire a year later. The renewal rate is still above 50% in this period, so a majority of expiring sectors re-lock. L/CS declines 13%, driven by both modest locked FIL contraction and circulating supply growth.
- **Second year (acceleration phase):** The cumulative effect of declining renewals becomes visible as sectors locked in the first year of the forecast begin to expire. With the renewal rate now approaching 36%, roughly two-thirds of expiring sectors release their pledge. Locked FIL drops 39% while circulating supply continues to grow (+15%) as released pledge and ongoing minting add to circulation. L/CS falls 47%, compounded by both the declining numerator and growing denominator.

SP Economics

Pledge per sector exhibits an inverse relationship with network size under attrition. As QAP contracts in the Decline scenario, the consensus pledge component — which scales with the ratio of sector size to total network power — increases from 0.17 to 0.36 FIL per 32 GiB sector, a 109% increase in collateral requirements despite a shrinking network.

The 1-year forward rate of return (FoFR) rises in the Decline scenario to approximately 50%, up from 23%. This is mechanical: fewer sectors compete for block rewards that decline more slowly than network power. This elevated return represents the equilibrium incentive for remaining providers but may be insufficient to attract new entrants if accompanied by adverse token price dynamics.

Key Takeaways

1. **Even at flat conditions, the network is slowly contracting.** At current onboarding and renewal rates, locked FIL declines 2% and L/CS falls from 0.122 to 0.108 over two years. The Status Quo scenario is not a growth scenario — it is a slow-erosion scenario.
2. **The Decline scenario is the more realistic projection.** Raw byte onboarding has already declined ~3x over the past year. The Decline scenario extrapolates this trend and projects locked FIL falling 39% and L/CS compressing to 0.065 over two years.
3. **Locked FIL has significant inertia.** The 360-day sector duration creates a lag between declining conditions and their effect on locked FIL. In the Decline scenario, locked FIL is only down 5% after one year despite QAP already contracting 12%. The full impact materializes in the second year as sectors locked during the transition begin to expire without renewal.
4. **Both onboarding and renewal rate drive the decline, but renewals dominate the flow.** The daily volume of power flowing through renewals (expiring sectors being re-locked) is roughly 10x larger than new onboarding in power terms. Changes in renewal behavior therefore have an outsized effect on locked FIL relative to changes in new onboarding.
5. **Attrition is self-reinforcing in pledge terms.** As the network shrinks, pledge per sector rises (+109% in the Decline scenario), which may further discourage participation. FoFR also rises as a partial offset, but the net effect on SP decision-making depends on token price dynamics not modeled here.
6. **Circulating supply is bounded.** Even under the Decline scenario, circulating supply increases by only ~15% over two years (837M to 960M FIL). The supply trajectory is dominated by vesting and minting schedules, with pledge dynamics providing a secondary modulating effect.

Appendix: Model Notes

- **Data source:** Spacescope API (on-chain supply and power statistics)

- **History window:** 365 days (Feb 2025 – Feb 2026)
- **Forecast horizon:** 730 days (Feb 2026 – Feb 2028)
- **Renewal rate:** 30-day mean (72.1%). The mean is used rather than the median (82.1%) because locked FIL is a cumulative stock: $\text{total pledge retained} = \sum_i \text{RR}_i \times \text{scheduled_expiration}_i \approx \overline{\text{RR}} \times \sum_i \text{scheduled_expiration}_i$. The mean captures the full distribution including low-RR days that the median would discard.
- **FIL+ rate:** Held constant at 97.1% across all scenarios.
- **Sector duration:** 360 days (average).
- **Initialization:** Pledge/reward split estimated from 180-day linear vesting integral of on-chain minting data (4.5% reward-locked, 95.5% pledge-locked).
- **Reward vesting approximation:** The model uses a balance/180 decay for reward release rather than per-cohort linear vesting. This introduces approximately 2–3M FIL of drift over the historical window, which is acceptable for forecasting purposes.